

Data Curation Report

Project Title: Supply Chain Resiliency & Margin Protection in Prestige Beauty: An Enterprise Integration Case Study

Date: June 12, 2026

Step 1 — Data Sourcing

Three data sources are integrated to comprehensively model the dual-sourcing transition problem: primary operational logs, secondary consumer sell-out sentiment, and external trade policy benchmarks.

Dataset 1 (Primary): Fashion & Beauty Supply Chain — Makeup Products

Field	Detail
File Name	supply_chain_data.csv
Source	https://www.kaggle.com/datasets/harshsingh2209/supply-chain-analysis
License	CC0 Public Domain — unrestricted use
Size	20.6 KB 100 Rows & 24 Columns
How Obtained	Direct download from Kaggle
Description	Operational supply chain data collected from a real fashion and beauty startup. Captures procurement, logistics, and manufacturing metrics across 100 SKUs in skincare (40%), haircare (34%), and cosmetics (26%). Serves as the primary master data transactional history for sourcing bottleneck mapping, cost-accounting calculations, and tariff scenario modeling.

Dataset 2 (Secondary): Sephora Products and Customer Reviews

Field	Detail
File Names	product_info.csv, reviews_0-250.csv, reviews_250-500.csv, reviews_500-750.csv, reviews_750-1250.csv, reviews_1250-end.csv
Source	https://www.kaggle.com/datasets/nadyinky/sephora-products-and-skincare-reviews
License	Public use via Kaggle
Size	Product file: 7.5 MB Reviews (combined): 496.9 MB 1,094,411 Total Rows
How Obtained	Direct download from Kaggle (6 files)
Description	Real product catalog and consumer review data scraped from Sephora.com. Covers prestige and luxury brands including Dior, Lancôme, Clinique, and TOM FORD across Skincare, Makeup, and Hair. Review timestamps spanning 2008 to 2023 enable time-series analysis of consumer sell-out momentum.

Dataset 3 (External Trade Benchmarks): USTR Section 301 & EU Commission Chapter 33

Field	Detail
Sources	USTR Section 301 Tariff Actions: https://ustr.gov/issue-areas/enforcement/section-301-investigations/section-301-china/tariff-actions EU Commission MFN Tariff Rates HS Chapter 33: https://taxation-customs.ec.europa.eu/customs-4/calculation-customs-duties/customs-tariff_en

License	Public domain — U.S. and EU government publications
How Obtained	Direct reference from government publications — no file download required. Tariff rates injected as engineered parameter values during data curation.
Description	Published tariff schedules for beauty sector HS codes 3303–3307. Used to engineer the Tariff Impact Coefficient — a category-level proxy mapping macro trade penalties directly to Product type in Dataset 1: cosmetics → 25% penalty; skincare → 10% premium; haircare → 0% near-shore baseline. This bypasses geographic limitations without introducing data scope creep.

Step 2 — Dataset Profiling

Data profiling conducted using Python (pandas) with .info(), .describe(), .nunique(), and .isnull().sum() functions. All statistics below are derived from actual data execution on the uploaded datasets.

Dataset 1: Supply Chain Data — Profile

Structure: 100 rows, 24 columns, 20.6 KB. Zero null values across all 24 fields.

- Categories: Skincare (40 SKUs), haircare (34 SKUs), cosmetics (26 SKUs)
- Suppliers: 5 anonymized vendors (Supplier 1: 27, Supplier 2: 22, Supplier 5: 18, Supplier 4: 18, Supplier 3: 15)
- Logistics Network: 4 transportation modes: Road (29%), Rail (28%), Air (26%), Sea (17%) distributed across 3 routes
- Inspection results: Pending (41%), Fail (36%), Pass (23%) — 36% failure rate flagged as key analytical finding

Numeric Ranges: Price (\$1.70 – \$99.17) | Manufacturing costs (\$1.09 – \$99.47) | Supplier Lead times (1 – 30 days) | Defect rates (0.02 – 4.94, avg 2.28) | Total landed cost (\$103.92 – \$997.41)

Dataset 2a & 2b: Sephora Catalog & Reviews — Profile

Catalog Structure: 8,494 rows, 27 columns. Price range: \$3.00 – \$1,900.00. Avg rating: 4.19. Reviews: 1 – 21,281 (avg 448).

Reviews Structure: 1,094,411 rows across 5 files, 19 columns. Date range: August 28, 2008 to March 21, 2023.

Data Quality Findings:

- Catalog missing critical data in value_price_usd (94.7% null) and sale_price_usd (96.8% null) — both dropped
- Reviews missing data in helpfulness (55.1% null) — dropped
- Additional catalog nulls: variation_desc (85.3%), child_max/min_price (67.6%), highlights (26.0%), size (19.2%), ingredients (11.1%)
- Reviews nulls retained for analysis: is_recommended (19.5%), skin_tone (17.6%), skin_type (12.4%)

Relationship Discovery

Operational Integration: Dataset 1 (Operational Logs) utilizes generic identifiers (SKU0 to SKU99) and cannot be directly joined via a simple text string to Dataset 2a (Sephora Catalog).

Data Bridge Logic: To pass forward-looking demand signals down to operational stocking tiers, the datasets are linked at the category level. The operational Product type column maps directly to the Sephora primary_category column using a structural mapping table.

Relational Keys: Dataset 2a (Product Info) connects directly to Dataset 2b (Reviews) using an exact string match on the product_id foreign key field.

Step 3 — Data Wrangling

All wrangling executed in Python (pandas) in Jupyter Notebook. Every decision below is documented with business rationale to ensure the cleaning process does not corrupt the analytical integrity of the data.

Dataset 1: Supply Chain Data — Wrangling

Tool: Python (pandas) | Zero null values: No null treatment required.

Column	Nulls	Action	Reason
Lead times	0	Rename → supplier_lead_time	Represents days from PO placement to supplier dispatch. Maps to SAP S/4HANA planned delivery parameters.
Lead time	0	Rename → delivery_lead_time	Represents total order-to-delivery days. The 9.9-day variance between these two fields exposes significant buffer stock risk.
Costs	0	Rename → total_landed_cost	Values (\$103–\$997) exceed sum of manufacturing and shipping. Represents total landed cost including overhead, which protects gross margin calculations.
Inspection results	0	Retain as-is	36% Fail rate is a primary analytical finding. Pending (41%) treated as incomplete. Both retained for EDA defect concentration analysis.
Customer demographics	0	Retain as-is	'Unknown' (31%) likely represents B2B or untracked channel — retained as valid category.

Outlier Risk Management

Manufacturing costs peak at a maximum of \$99.47 against a \$47.27 average. This high-cost SKU represents the primary point of margin risk exposure. When the 25% tariff penalty is applied, its cost balloons by \$24.87, making it the top priority for near-shore dual-sourcing substitution.

Feature Engineering — Dataset 1:

- Tariff_Impact_Coefficient:** Engineered variable mapping Product type to 2026 tariff tier — cosmetics: 25% Section 301 East Asia penalty; skincare: 10% EU MFN premium; haircare: 0% near-shore baseline. Applied to Manufacturing costs to generate Tariff_Adjusted_Mfg_Cost. Functions as dynamic toggle variable in Python pipeline and Tableau dashboard.
- Tariff_Adjusted_Mfg_Cost:** manufacturing_costs × (1 + tariff_impact_coefficient). Enables scenario modeling of COGS inflation under current 2026 trade conditions.

- **lead_time_variance:** $| \text{supplier_lead_time} - \text{delivery_lead_time} |$. Quantifies supply chain planning risk per SKU. Avg 9.9 days; max 28 days.
- **revenue_per_unit:** $\text{revenue_generated} \div \text{number_of_products_sold}$. Enables per-SKU yield analysis.
- **stock_to_sales_ratio:** $\text{stock_levels} \div \text{number_of_products_sold}$. Proxy for inventory health.

Dataset 2a & 2b: Sephora Data — Wrangling

Dropped Fields: Removed variation_desc, value_price_usd, sale_price_usd, and child_max/min_price due to extreme null rates (>67%). Removed helpfulness, eye_color, and hair_color as not relevant to supply chain operations.

Retained & Filled Fields: Retained size, ingredients, and tertiary_category. Replaced missing data with a standard 'Not Listed' placeholder to preserve rows for volume analysis.

Feature Engineering: Extracted review_year and review_month from submission_time timestamp to turn historical text reviews into a forward-looking S&OP demand sensing signal.

Cleaned Enterprise Schemas (Standardized snake_case)

Schema 1: Supply Chain Data (Cleaned Master Payload)

100 rows | 27 columns | 20.6 KB | Source: Kaggle CCO

Technical Field Name	Data Type	Business Description
product_type	STRING	Product category: skincare, haircare, or cosmetics. Primary Tariff Impact Coefficient mapping key.
sku	STRING	Unique product identifier (SKU0–SKU99). Primary master key.
price	FLOAT	Unit selling price in USD (\$1.70–\$99.17).
availability	INTEGER	Current warehouse units available on a 0–100 scale.
number_of_products_sold	INTEGER	Total unit sales volume accumulated per SKU.
revenue_generated	FLOAT	Total revenue per SKU in USD (\$1,061–\$9,866).
customer_demographics	STRING	Primary channel customer segment.
stock_levels	INTEGER	Current warehouse physical stock count.
order_quantities	INTEGER	Units ordered per batch procurement purchase order.
shipping_times	INTEGER	Transit days from supplier node to distribution center.
shipping_carriers	STRING	Anonymized carrier provider.
shipping_costs	FLOAT	Per-unit freight cost allocation.
supplier_name	STRING	Anonymized production vendor node (Supplier 1–5).
location	STRING	Manufacturing supplier city node.
supplier_lead_time	INTEGER	Days from purchase order placement to vendor dispatch.
delivery_lead_time	INTEGER	Total order-to-delivery turnaround days.
production_volumes	INTEGER	Total batch sizes processed per manufacturing run.

manufacturing_lead_time	INTEGER	Production cycle time required within the factory.
manufacturing_costs	FLOAT	Raw baseline manufacturing cost before trade penalties.
inspection_results	STRING	Quality check outcome: Pass, Fail, or Pending.
defect_rates	FLOAT	Defective parts per hundred produced per SKU.
transportation_modes	STRING	Logistics transit mode: Air, Rail, Road, or Sea.
routes	STRING	Anonymized geographic transit route.
total_landed_cost	FLOAT	Total landed cost per SKU including operational overhead.
tariff_impact_coefficient	FLOAT	Category tariff proxy: cosmetics=0.25, skincare=0.10, haircare=0.00.
tariff_adjusted_mfg_cost	FLOAT	Calculated field: $\text{manufacturing_costs} \times (1 + \text{tariff_impact_coefficient})$.
lead_time_variance	INTEGER	Calculated planning risk buffer: $ \text{supplier_lead_time} - \text{delivery_lead_time} $.
revenue_per_unit	FLOAT	Unit pricing yield: $\text{revenue_generated} \div \text{number_of_products_sold}$.
stock_to_sales_ratio	FLOAT	Inventory velocity: $\text{stock_levels} \div \text{number_of_products_sold}$.

Schema 2a: Sephora Product Info (Cleaned Catalog)

8,494 rows | 21 columns | 7,727 KB | Source: Sephora.com via Kaggle

Technical Field Name	Data Type	Business Description
product_id	STRING	Unique product alphanumeric key. Relational join field.
product_name	STRING	Complete consumer product name on retail channels.
brand_id	INTEGER	Numeric enterprise manufacturer index.
brand_name	STRING	Market brand identifier.
loves_count	INTEGER	Total consumer digital wish-list saves. Demand signal indicator.
rating	FLOAT	Average item rating out of 5.0 stars.
reviews	INTEGER	Total customer review submission counts.
size	STRING	Unit volume sizing footprint.
ingredients	STRING	Complete ingredient list used for raw material validation.
price_usd	FLOAT	Active retail shelf price in USD.
out_of_stock	BOOLEAN	Stock status flag. Tracks real-time shelf depletion.
primary_category	STRING	Top-level inventory category. Joins with operational files.
price_tier	STRING	Engineered pricing bucket: Under \$25, \$25–\$50, \$50–\$100, \$100–\$200, \$200+.
variation_type	STRING	Type of product variation (e.g., Shade, Size). 'Not Listed' where null.
variation_value	STRING	Value of product variation. 'Not Listed' where null.
limited_edition	BOOLEAN	1 = limited edition (7% of products); 0 = standard.
new	BOOLEAN	1 = newly listed product (7.2%); 0 = standard.

online_only	BOOLEAN	1 = available online only (21.9%); 0 = sold in-store.
sephora_exclusive	BOOLEAN	1 = Sephora exclusive product (27.9%); 0 = multi-retailer.
secondary_category	STRING	Sub-category (e.g., Moisturizers, Eye, Face).
tertiary_category	STRING	Detailed sub-category. 'Not Listed' where null (11.7% fill).

Schema 2b: Sephora Reviews (Cleaned Sentiment Logs)

1,094,411 rows | 15 columns | ~490 MB combined | Source: Sephora.com via Kaggle

Technical Field Name	Data Type	Business Description
author_id	INTEGER	Anonymized unique customer reviewer profile key.
rating	INTEGER	Individual star rating awarded by reviewer (1–5).
is_recommended	FLOAT	Direct consumer recommendation vote.
submission_time	DATE	Exact calendar date of review publication.
review_text	STRING	Complete customer review feedback narrative.
product_id	STRING	Product identifier. Links back to Schema 2a.
review_year	INTEGER	Engineered calendar year field. Used for time-series velocity tracking.
review_month	INTEGER	Engineered calendar month field. Captures seasonal demand spikes.
total_feedback_count	INTEGER	Total helpful/unhelpful votes received on review.
total_neg_feedback_count	INTEGER	Number of 'not helpful' votes.
total_pos_feedback_count	INTEGER	Number of 'helpful' votes.
review_title	STRING	Review headline where provided (27.7% null).
skin_tone	STRING	Reviewer self-reported skin tone (17.6% null).
skin_type	STRING	Reviewer self-reported skin type: combination, dry, normal, oily (12.4% null).
brand_name	STRING	Brand name at time of review.
price_usd	FLOAT	Product price at time of review.

Generative AI Disclosure

Claude (Anthropic, Claude Sonnet 4.6) via Claude.ai was utilized as a documentation and technical partner for this deliverable:

- **Dataset profiling execution:** Claude ran Python scripts directly on uploaded datasets to produce real distribution statistics, null counts, and data quality findings. All numerical values in this document are derived from actual data execution and not AI-generated estimates.
- **Anomaly identification:** Claude assisted in identifying the duplicate lead time columns, the 36% inspection failure rate, and the Costs field interpretation. All reviewed and confirmed by the author using 20+ years of ERP and supply chain domain expertise.

- **Document structure and schema creation:** Claude assisted in formatting wrangling decision tables and data table schemas. All drop/retain decisions and business rationale reflect the author's analytical judgment.
- **Tariff Impact Coefficient validation:** Claude stress-tested the logic of mapping tariff liabilities to product category rather than geography, confirming the Structural Attribute Proxy Strategy as analytically valid given the dataset's geographic constraints.